Model Evaluation — ML-Based Traffic Speed Prediction

This document describes the evaluation of the trained models published in this repository (`paper_artifacts/trained_models_ba` accompanying the paper "Cloud-Fog-Edge Deployment of ML-Based Traffic Speed Prediction: An Empirical Analysis of Performance, Cost, and Scalability".

Models

One multilayer perceptron per sensor direction (37 sensor directions; layer sizes 96-256-256-96-256, ReLU with dropout, ~142k parameters, RMSprop optimiser, MSE loss, float32). Each model ships with its fitted input (x) and output (y) RobustScaler as `.pkl` files. Models were trained on the initial three-year window (1 December 2020 to 1 December 2023) of five-minute average-speed records from Greater Manchester drakewell sensors, using 15 engineered features (cyclic time encodings and event flags). These are the base models from which all warm-start re-training scenarios in the paper are initialised.

Evaluation protocol

Each sensor's data is split chronologically 80/20; the final 20% of the training window serves as the held-out test set (approximately the last seven months of 2023 for the three-year window). Metrics: mean absolute error (MAE, mph), root mean squared error (RMSE, mph), and mean absolute percentage error (MAPE, %). A complementary common-test-set evaluation (identical held-out periods for all training scenarios) is summarised in the last section.

Per-sensor results (held-out test set)

#	Sensor	MAE (mph)	RMSE (mph)	MAPE (%)	Test samples
1	1020 nw	1.95	2.92	8.02	61,132
2	1020 se	1.67	2.42	6.48	60,958
3	1056 e	1.36	1.92	6.06	59,752
4	1056 w	1.75	2.47	7.57	60,544
5	1063 e	1.45	2.50	5.76	61,213
6	1063 w	1.59	2.47	5.50	61,253
7	1071 ne	2.09	3.65	10.48	62,524
8	1071 sw	1.31	1.93	4.31	62,668
9	1163 nw	2.37	3.36	9.55	62,835
10	1163 se	2.97	4.74	16.52	62,843
11	1165 se	2.07	3.04	8.16	61,874
12	1173 n	1.83	2.63	6.38	58,707
13	1273 n	1.90	2.99	7.41	61,176
14	1276 e	1.56	2.24	6.85	60,057
15	1276 w	3.52	5.50	27.63	59,959
16	1385 n	2.10	3.24	11.30	58,161
17	1404 e	2.69	4.38	11.19	61,257
18	1404 w	2.60	3.93	10.09	60,197
19	1407 n	1.90	3.20	10.03	62,645
20	1407 s	2.14	3.12	9.71	62,573
21	1413 ne	1.58	2.27	6.06	62,277
22	1413 sw	1.64	2.25	5.92	62,174
23	1414 ne	1.54	2.13	6.18	60,300
24	1414 sw	1.41	1.98	5.66	60,317
25	1417 s	1.47	2.08	7.07	62,237
26	1418 e	1.92	3.28	9.09	55,437
27	1418 w	1.74	3.06	7.90	55,694
28	1420 nw	2.34	3.59	10.31	62,082
29	1420 se	1.91	2.90	8.64	61,899
30	1425 e	1.80	2.48	3.36	62,208

#	Sensor	MAE (mph)	RMSE (mph)	MAPE (%)	Test samples
31	1425 w	2.16	4.38	6.25	62,376
32	1426 e	2.26	3.25	4.32	61,908
33	1426 w	3.75	6.26	12.68	62,685
34	1428 e	3.88	7.14	12.68	61,259
35	1428 w	2.17	3.25	4.12	61,158
36	1429 e	2.02	2.81	7.04	61,680
37	1429 w	2.00	2.90	7.04	60,874

Aggregate: mean MAE 2.07 mph (median 1.92); mean MAPE 8.47% (median 7.41%). The two highest-error sensor directions (platform 1404 east/west) are markedly more variable, frequently congested locations; per-sensor error correlates strongly with speed variability (Spearman rho = 0.79 with the coefficient of variation of speed) and congestion frequency (rho = 0.82).

Common-test-set scenario comparison

All training scenarios of the paper (3-year base, 4.2-year cold start, and progressive warm-start re-training windows PW3/PW6/PW12) were additionally evaluated on identical held-out periods. Mean MAE (mph) per evaluation window:

Scenario	Dec 2024 - Feb 2025	Oct 2024 - Feb 2025	May 2024 - Feb 2025
base (3y)	3.09	2.83	2.44
cold start (4.2y)	3.02	2.71	2.32
re-train PW3	2.91	2.68	2.35
re-train PW6	2.89	2.64	2.31
re-train PW12	2.90	2.61	excluded*

*PW12 trains up to September 2024, overlapping the nine-month window.

Warm-start re-training improves on both cold-start baselines in every window; prediction accuracy is invariant to the training platform (across repeated runs and platforms, the standard deviation of aggregate MAPE is below 0.3 percentage points in over 90% of measured cells).